

Supplementary Information — Advanced Machine Learning Strategies for Predicting Therapy Response in Preclinical Glioblastoma using Longitudinal MRI

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ABSTRACT

This document provides supplementary methods, tables, and figures supporting the main manuscript. Supplementary items are referenced in the main text as Supplementary Methods, Supplementary Tables S1–S4, and Supplementary Figures S1–S5.

Supplementary Methods

EfficientNetB0 architecture and feature extraction

EfficientNetB0 is a convolutional neural network from the EfficientNet family proposed by Tan and Le¹. Conventional convolutional architectures scale one dimension of the network at a time (depth, width, or input resolution). EfficientNet instead applies *compound scaling*, jointly scaling all three dimensions with a fixed set of coefficients, which yields a more favorable accuracy-to-parameter trade-off. The baseline EfficientNetB0 contains approximately 5.3 million parameters, substantially fewer than VGG16 (~138M), ResNet50 (~25M), or DenseNet121 (~8M), making it well-suited to small medical imaging datasets where over-parameterized models are prone to overfitting.

Two configurations were evaluated in this study:

- **Frozen feature extractor (FE).** EfficientNetB0 was used with all layers frozen. Each preprocessed input image was passed through the network followed by a global average pooling layer. A dense layer with ReLU activation, batch normalization, and dropout produced a 256-dimensional embedding vector encoding spatial and textural patterns from the MRI scans. These embeddings were then classified with an XGBoost model under the same Grouped Leave-One-Out (GLOO) cross-validation scheme as the radiomics pipeline, enabling a direct comparison.
- **Fine-tuned model (FT).** The EfficientNetB0 backbone was partially unfrozen (last 50 layers trainable) and equipped with a custom binary classification head (global pooling, dense layer, batch normalization, dropout, and a final sigmoid activation). The model accepted three-channel inputs and was trained end-to-end with a reduced learning rate to adapt to domain-specific features while preserving low-level ImageNet representations.

All images were converted to the TFRecord format and processed through a TensorFlow `tf.data` pipeline enabling on-the-fly augmentation, shuffling, batching, and prefetching. A batch size of 32 was used as a compromise between gradient stability and computational efficiency. Random oversampling of the minority (cured) class was applied only within the training partition of each fold to avoid information leakage.

XGBoost classifier

XGBoost² is an ensemble method based on gradient-boosted decision trees. It builds an additive model in a forward stage-wise manner, at each step fitting a new tree to the negative gradient of a differentiable loss function and using a second-order Taylor approximation of the loss for more precise optimization. Regularization is achieved through L1 and L2 penalties on the leaf weights, shrinkage (learning rate), column subsampling, and tree pruning, all of which reduce overfitting in small, high-dimensional datasets.

The final hyperparameter configuration is reported in the main manuscript (radiomics XGBoost classifier configuration table). The complete grid explored during optimization is provided in Supplementary Table S2. Class-imbalance handling strategies (SMOTE³, ENN⁴, and training on the original distribution) were compared, with controlled oversampling providing the best trade-off between sensitivity and specificity for the minority cured class.

Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM)⁵ produces class-discriminative localization maps by using the gradients of the target class score with respect to the feature maps of the last convolutional layer. For a target class c and feature maps A^k , the neuron importance weights are computed as

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \quad (S1)$$

and the localization map is obtained by a ReLU over the weighted combination of feature maps,

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right). \quad (S2)$$

The resulting heatmaps were upsampled to the input resolution and overlaid on the original MRI to identify the anatomical regions driving each prediction. Maps were generated for both correctly classified and misclassified examinations.

Radiomic feature families and image transformations

A total of 1,218 features were extracted per image using PyRadiomics⁶, following the Image Biomarker Standardization Initiative (IBSI) recommendations⁷. The feature families were:

- **First-order statistics:** mean, variance, entropy, skewness, kurtosis, energy, and related intensity descriptors.
- **Shape-based descriptors:** elongation, compactness, sphericity, and surface-area-related measurements, independent of intensity.
- **Gray-Level Co-occurrence Matrix (GLCM):** second-order texture features quantifying spatial relationships between neighboring voxel intensities.
- **Gray-Level Run-Length Matrix (GLRLM):** distribution of consecutive runs of similar gray levels.
- **Gray-Level Size-Zone Matrix (GLSZM):** homogeneous intensity zones capturing regional heterogeneity.
- **Gray-Level Dependence Matrix (GLDM):** local gray-level dependencies and neighborhood homogeneity.
- **Neighboring Gray-Tone Difference Matrix (NGTDM):** coarseness, complexity, and contrast.

Features were computed on the original images and on multiple transformed representations: Laplacian-of-Gaussian (LoG) filtered images ($\sigma = 1.0, 2.0, 3.0$), one-level wavelet decompositions (eight frequency-specific sub-bands), gradient images, and intensity transformations (square, square-root, logarithmic, and exponential). Temporal descriptors (first-order differences, rolling means and standard deviations, local slopes, and relative percentage changes) were subsequently derived from the retained features to encode longitudinal dynamics without modeling complete sequences as a jointly classified multivariate time series.

Supplementary Tables

Code	Group	Images per Exploration	Number of Explorations	Range of Days Studied
C1258	Control	1–3	6	7–24
C1260	Control	1–4	6	8–18
C1261	Control	1–5	8	10–26
C1297	Control	2	1	23
C1299	Control	2	1	17
C1359	Control	1–4	3	11–16
C1360	Control	1–4	6	11–22
C1361	Control	1–4	5	11–20
C1276	TMZ-Treated – Cured	1–2	15	11–45
C1281	TMZ-Treated – Cured	1–2	8	10–27
C1284	TMZ-Treated – Cured	1	8	10–27
C1285	TMZ-Treated – Cured	1–2	10	9–31
C1382	TMZ-Treated – Cured	1	8	11–29
C1263	TMZ-Treated – Relapsing	1–4	18	11–47
C1264	TMZ-Treated – Relapsing	1–4	16	11–39
C1270	TMZ-Treated – Relapsing	1–4	14	10–37
C1380	TMZ-Treated – Relapsing	1–4	17	11–41
C1383	TMZ-Treated – Relapsing	1–4	19	11–49

Table S1. Per-subject description of the dataset. Each row corresponds to one animal, identified by a unique code (CXXXX). Columns report the group, the number of MRI images acquired at each time point, the number of explorations (imaging sessions) performed over time, and the range of days studied, taking day 0 as the tumor inoculation day.

Hyperparameter	Values explored
max_depth	{2, 3, 4, 5}
learning_rate	{0.01, 0.05, 0.1}
n_estimators	{100, 200, 300}
min_child_weight	{1, 3, 5}
gamma	{0, 0.5, 1}
subsample	{0.6, 0.8, 1.0}
colsample_bytree	{0.6, 0.8, 1.0}
Resampling	{none, SMOTE, ENN}

Table S2. Hyperparameter search grid for the radiomics-based XGBoost classifier. The final configuration selected by cross-validation is reported in Supplementary Table S3.

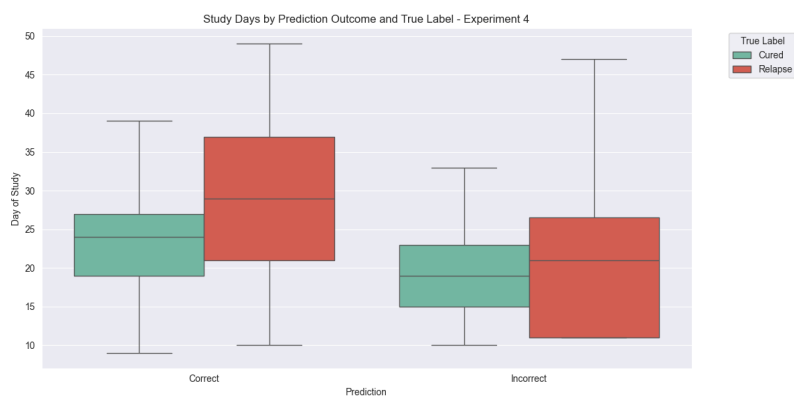
Parameter	Value
Base learner	Gradient-boosted decision trees
max_depth	3
learning_rate	0.01
n_estimators	100
subsample	0.6
colsample_bytree	0.6
gamma	0.5
Regularization	L1 and L2 penalties
Class-imbalance handling	SMOTE / ENN / none (compared)
Hyperparameter search	Grid search (see Supplementary Table S2)
Validation	Grouped Leave-One-Out (GLOO)

Table S3. Final configuration of the radiomics-based XGBoost classifier.

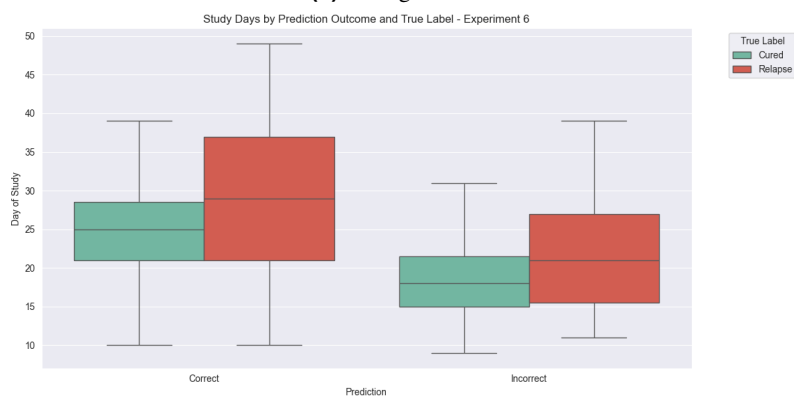
Parameter	Value
Backbone	EfficientNetB0 (ImageNet pre-trained)
Trainable layers	Last 50 layers unfrozen
Input	3-channel images
Classification head	Global pooling + dense + batch norm + dropout + sigmoid
Loss function	Binary cross-entropy
Optimizer	Adam
Batch size	32
Class-imbalance handling	Random oversampling (training folds only)
Regularization	Dropout, early stopping
Validation	Grouped Leave-One-Out (GLOO)

Table S4. Configuration of the EfficientNetB0 fine-tuning procedure.

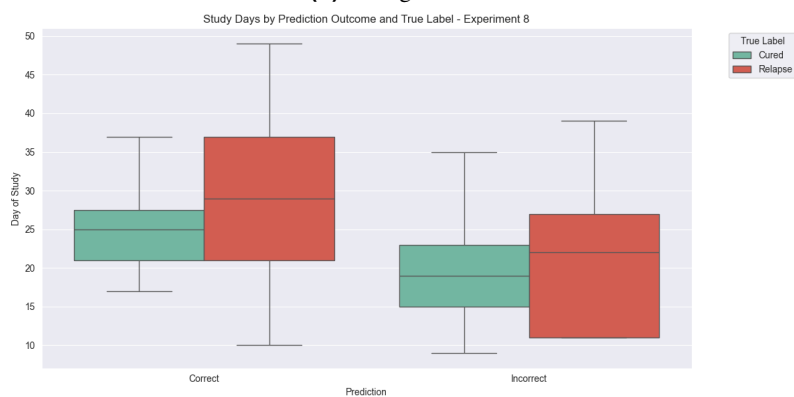
Supplementary Figures



(a) Configuration 4



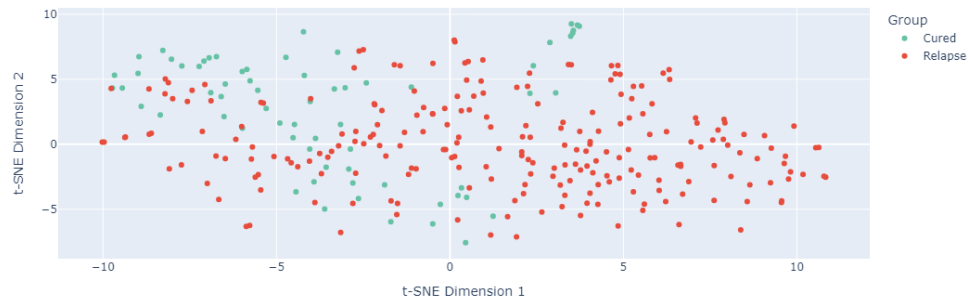
(b) Configuration 6



(c) Configuration 8

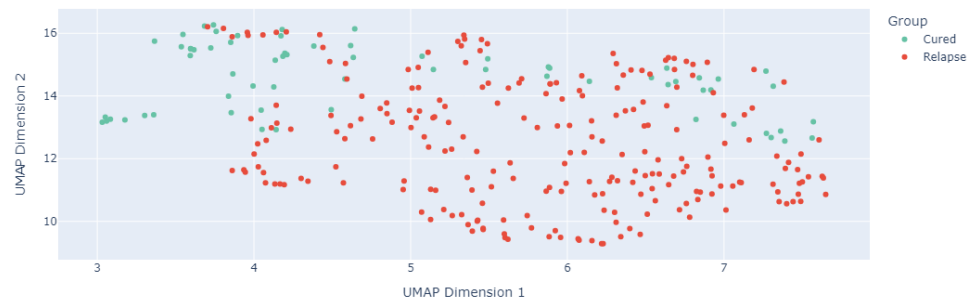
Figure S1. Distribution of cross-validation performance for representative radiomics configurations, illustrating the effect of the different class-imbalance handling strategies (original distribution, SMOTE, and ENN) on the XGBoost classifier.

2D Visualisation of Latent Space with t-SNE



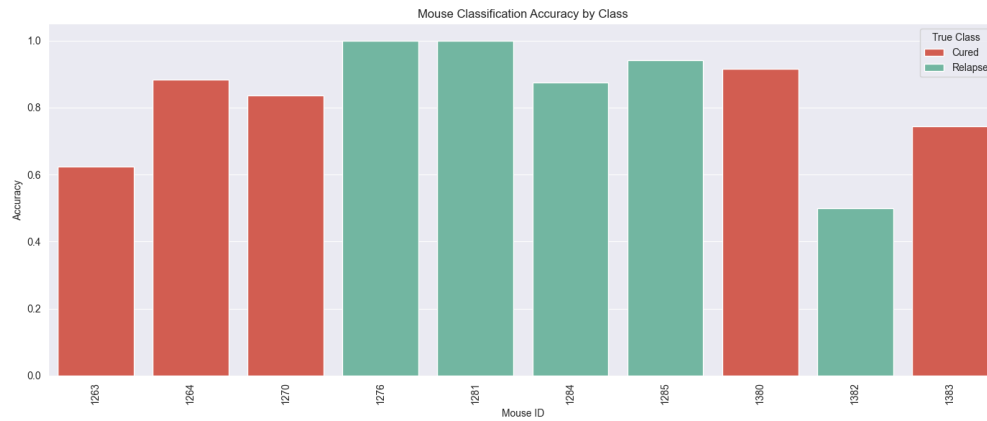
(a) t-SNE projection

2D Visualisation of Latent Space with UMAP

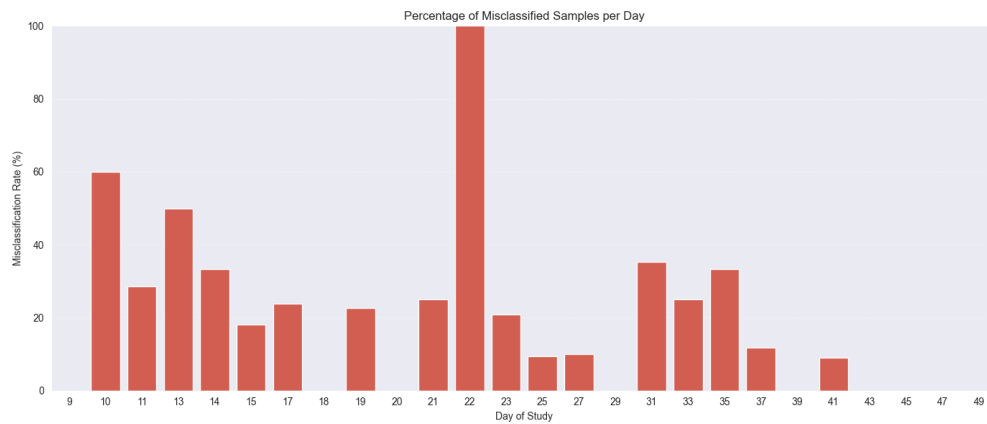


(b) UMAP projection

Figure S2. Low-dimensional projections (t-SNE and UMAP) of the selected radiomic feature space, colored by outcome group. These visualizations illustrate the partial separability between cured and relapsing examinations.

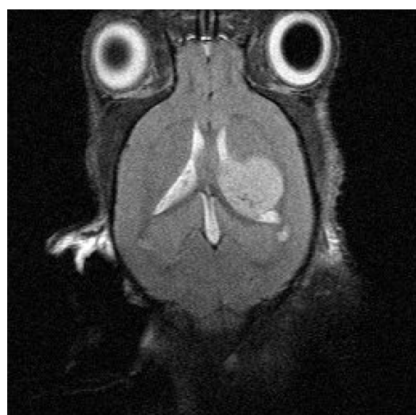


(a) Classification accuracy by class

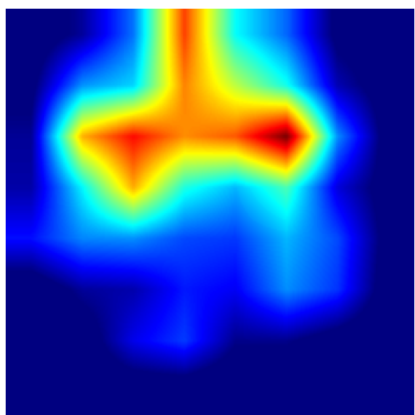


(b) Misclassified examinations per day

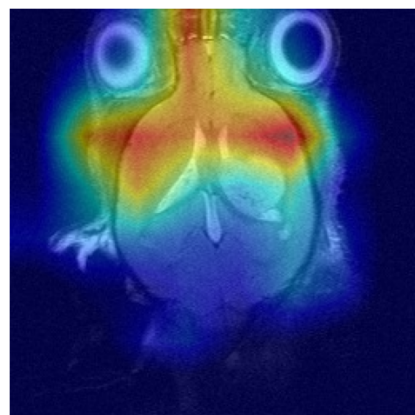
Figure S3. Temporal analysis of predictive performance for the deep learning model. (a) Classification accuracy per class and (b) number of misclassified examinations as a function of tumor evolution day, showing reduced performance at the earliest time points.



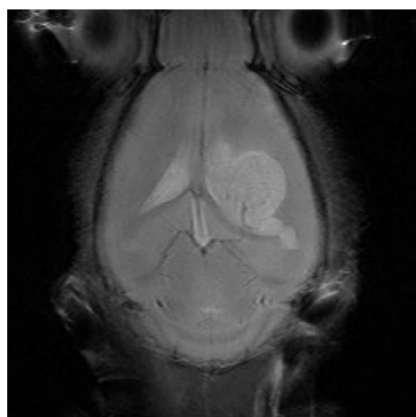
(a) Day 14 – MRI



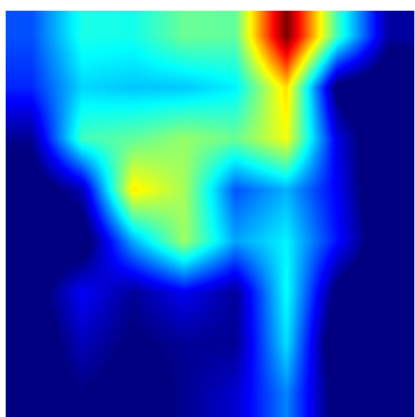
(b) Day 14 – Heatmap



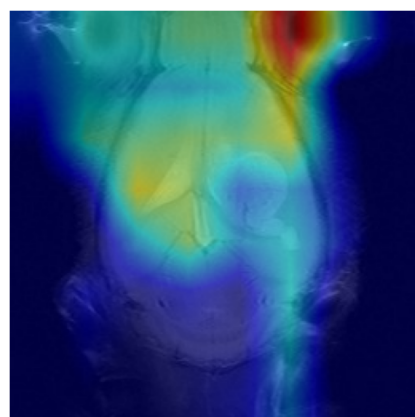
(c) Day 14 – Grad-CAM



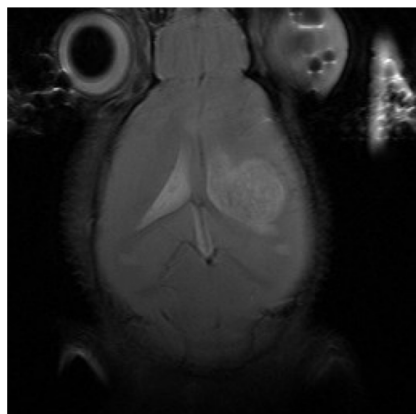
(d) Day 17 – MRI



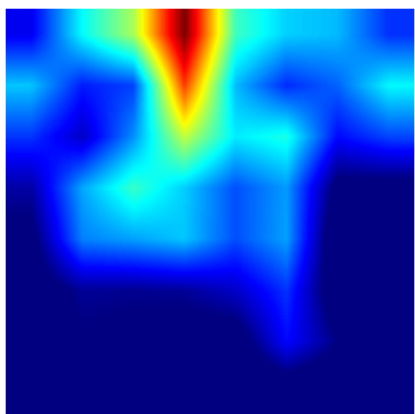
(e) Day 17 – Heatmap



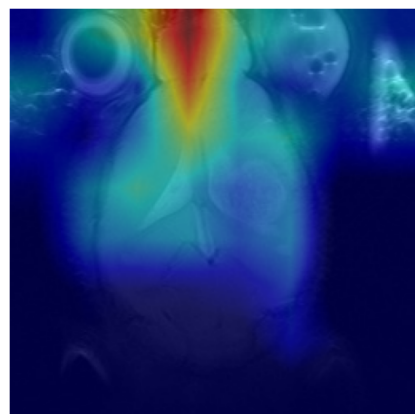
(f) Day 17 – Grad-CAM



(g) Day 19 – MRI



(h) Day 19 – Heatmap



(i) Day 19 – Grad-CAM

Figure S4. Temporal evolution of Grad-CAM attention maps for subject C1263 across days 14, 17, and 19, complementing the day-11 visualization shown in the main manuscript. Each row shows the original MRI, the Grad-CAM heatmap, and the overlay.

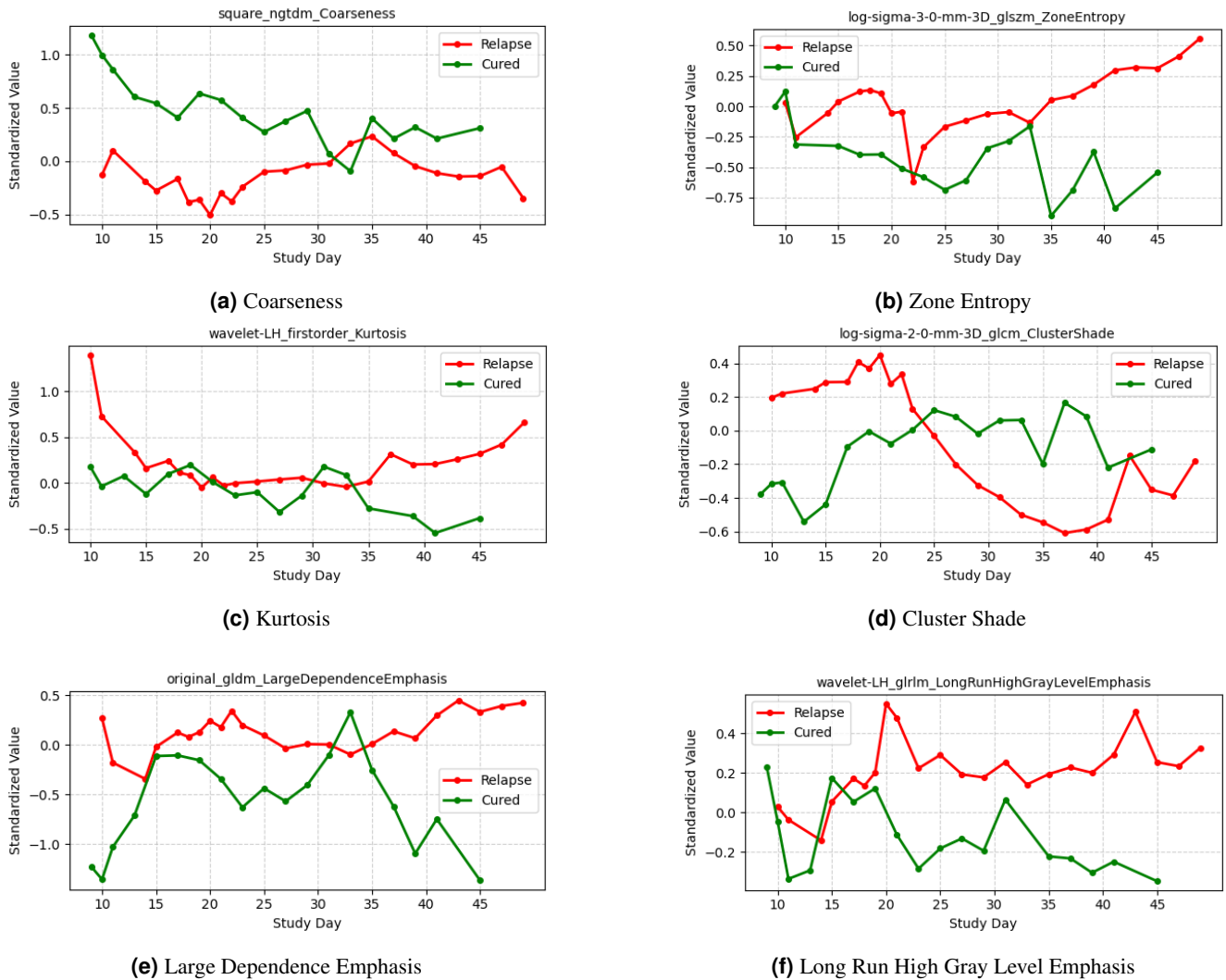


Figure S5. Temporal evolution of additional selected radiomic features across cured and relapsing groups, complementing the Elongation and Dependence Entropy trajectories shown in the main text.

References

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